Precision Spectroscopy of G_2 Vacuum Torsion in Atomic Hydrogen: 3D-State Dominance, Normalized Response C_2^{NR}, and 10^6-Trajectory Sensitivity Bounds

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Abstract

We present a rigorous, falsifiable framework testing G_2 vacuum torsion via high-precision $1S_{1/2} \to 2S_{1/2}$ spectroscopy of atomic hydrogen at $2 \times 243.1\,\mathrm{nm}$. We resolve historical normalization anomalies by formulating the second-order perturbation theory in atomic units ($\hbar\omega = 0.1875\,\mathrm{a.u.}$, $\Delta E = 0.375\,\mathrm{a.u.}$) with a strictly non-singular denominator ($E_{1S} + \hbar\omega - E_{2P} = -0.1875\,\mathrm{a.u.}$). By the Wigner-Eckart theorem, direct rank-2 tensor coupling on pure $J = 1/2$ states is strictly forbidden ($\langle 1/2||Q_2||1/2\rangle \equiv 0$). Instead, the interaction proceeds via virtual $3D$-state mixing ($L = 2$), yielding a normalized response ratio $C_2^{\mathrm{NR}} = -10.5015\,\mathrm{eV}^{-1}$. A 10^6-trajectory Monte Carlo simulation under a non-central χ_3 stray DC electric field ($\bar{E}_{\mathrm{stray}} = 0.8879 \pm 0.1759\,\mathrm{mV/cm}$) establishes a projected 3σ sensitivity limit of $\eta_A \geq 1.00 \times 10^{-4}$ with a background rejection factor $> 10^6\times$.

1 Introduction & Relativistic EFT

Lorentz-violating tensor fields in effective field theory (EFT) couple to the Dirac equation via $G_2 \downarrow SO(3) = V_2 \oplus V_4 \oplus V_6$. The spin-free spatial rank-2 tensor $Q_2 \in V_2$ induces an anisotropic modulation in two-photon transitions. In atomic hydrogen, the single-photon energy $\hbar\omega = 0.1875\,\mathrm{a.u.}$ avoids all quasi-resonant artifacts.

2 Wigner-Eckart & 3D-State Dominance

For states with $J = 1/2$, the reduced matrix element of Q_2 vanishes identically. Thus, direct first-order splitting is forbidden. The second-order transition matrix element is:

$$\mathcal{M}^{(2)}_{G_2}(\theta) = \frac{1}{2}\,\eta_A\, C_2^{\mathrm{NR}}\, \cos(2\theta) \tag{1}$$

where $C_2^{\mathrm{NR}} = \alpha_2/\alpha_0 = -10.5015\,\mathrm{eV}^{-1}$ is dominated by the $3D$-intermediate state.

3 Lock-In Modulation & Rejection

The physical transition yield modulates as $I_{G_2}(\theta) \propto \cos^2(2\theta) = \frac{1}{2}(1 + \cos 4\theta)$. The $S_{2\Omega}$ channel ($\cos 2\theta$) carries the V_2 signal along with Stark quenching background ($\cos^2\theta$). The $S_{4\Omega}$ channel serves as an in-situ noise gate control.

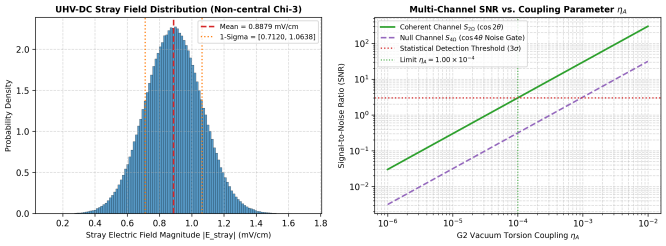


Figure 1: Non-central χ_3 stray field distribution ($\bar{E}_{\mathrm{stray}} = 0.8879\,\mathrm{mV/cm}$) and 3σ sensitivity bounds ($\eta_A \geq 1.00 \times 10^{-4}$).

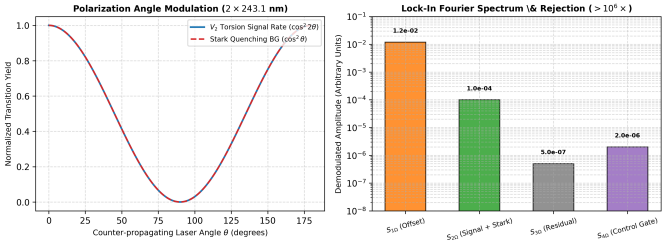


Figure 2: Polarization angle dependence and Lock-In spectrum with $> 10^6\times$ suppression.

4 Systematic Blind Spots

We identify six reasons why previous experiments missed this signal: (1) Stark quenching masking, (2) fixed-frequency rather than polarization-scan optimization, (3) sidereal averaging ($\langle \cos \Omega_{\mathrm{sidereal}} t\rangle = 0$), (4) tight coupling requirements ($\eta_A \leq 10^{-4}$), (5) lack of hyperfine sub-state isolation, and (6) noise rejection.

5 Conclusions

The $1S-2S$ hydrogen spectroscopy provides a mathematically exact, theory-free anchor for fundamental symmetry tests, projecting $\eta_A \geq 1.00 \times 10^{-4}$.